▶ IMMUNOLOGY—CELLULAR COMPONENTS

Innate vs adaptive immunity

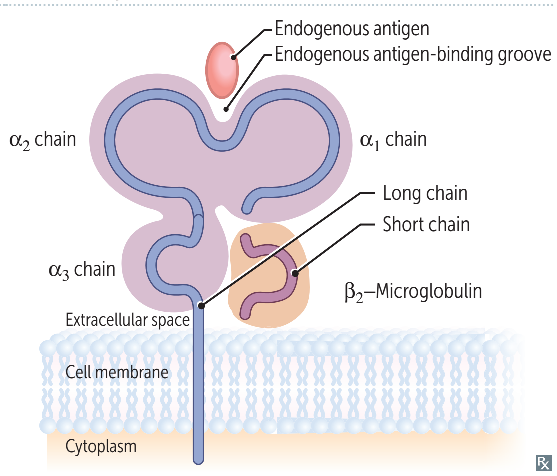
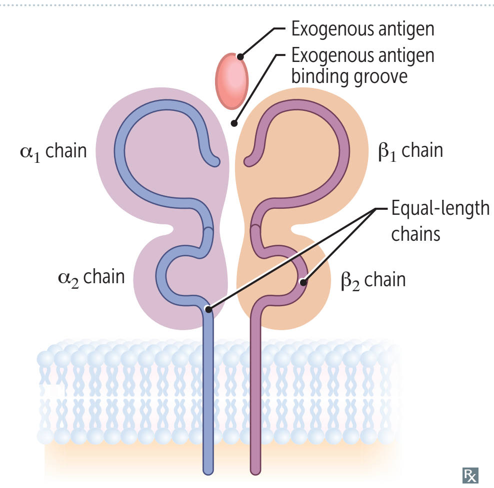
	Innate immunity	Adaptive immunity
COMPONENTS	Neutrophils, macrophages, monocytes, dendritic cells, natural killer (NK) cells (lymphoid origin), complement, physical epithelial barriers, secreted enzymes.	T cells, B cells, circulating antibodies.
MECHANISM	Germline encoded.	Variation through V(D)J recombination during lymphocyte development.
RESPONSE TO PATHOGENS	Nonspecific. Occurs rapidly (minutes to hours). No memory response.	Highly specific, refined over time. Develops over long periods; memory response is faster and more robust.
SECRETED PROTEINS	Lysozyme, complement, C-reactive protein (CRP), defensins, cytokines.	Immunoglobulins, cytokines.
KEY FEATURES IN PATHOGEN RECOGNITION	Toll-like receptors (TLRs): pattern recognition receptors that recognize pathogen- and damage-associated molecular patterns (PAMPs and DAMPs) → activation of NF-κB → release of pro-inflammatory cytokines. Examples of PAMPs: LPS (gram ⊖ bacteria), flagellin (bacteria), nucleic acids (viruses). Examples of DAMPs: mitochondrial DNA, histones, heat shock proteins.	Memory cells: activated B and T cells; subsequent exposure to a previously encountered antigen → stronger, quicker immune response. Adaptive immune responses decrease with age (immunosenescence).

Immune privilege

Organs (eg, eye, brain, placenta, testes) and tissues where chemical or physical mechanisms limit immune responses to foreign antigens to avoid damage that would occur from inflammatory sequelae. Allograft rejection at these sites is less likely.

Major histocompatibility complex I and II

MHC encoded by HLA genes. Present antigen fragments to T cells and bind T-cell receptors (TCRs).

	MHC I	MHC II
LOCI	HLA-A, HLA-B, HLA-C MHC I loci have 1 letter	HLA-DP, HLA-DQ, HLA-DR MHC II loci have 2 letters
BINDING	TCR and CD8 (CD8 × MHC 1 = 8)	TCR and CD4 (CD4 × MHC 2 = 8)
STRUCTURE	1 long chain, 1 short chain (3 α, 1 β)	2 equal-length chains (2 α, 2 β)
EXPRESSION	All nucleated cells, APCs, platelets (except RBCs)	APCs
FUNCTION	Present endogenous antigens (eg, viral or cytosolic proteins) to CD8+ cytotoxic T cells	Present exogenous antigens (eg, bacterial proteins) to CD4+ helper T cells
ANTIGEN LOADING	Antigen peptides loaded onto MHC I in RER after delivery via TAP (transporter associated with antigen processing)	Antigen loaded following release of invariant chain in an acidified endosome
ASSOCIATED PROTEINS	β_2-microglobulin	Invariant chain
STRUCTURE		

HLA subtypes associated with diseases

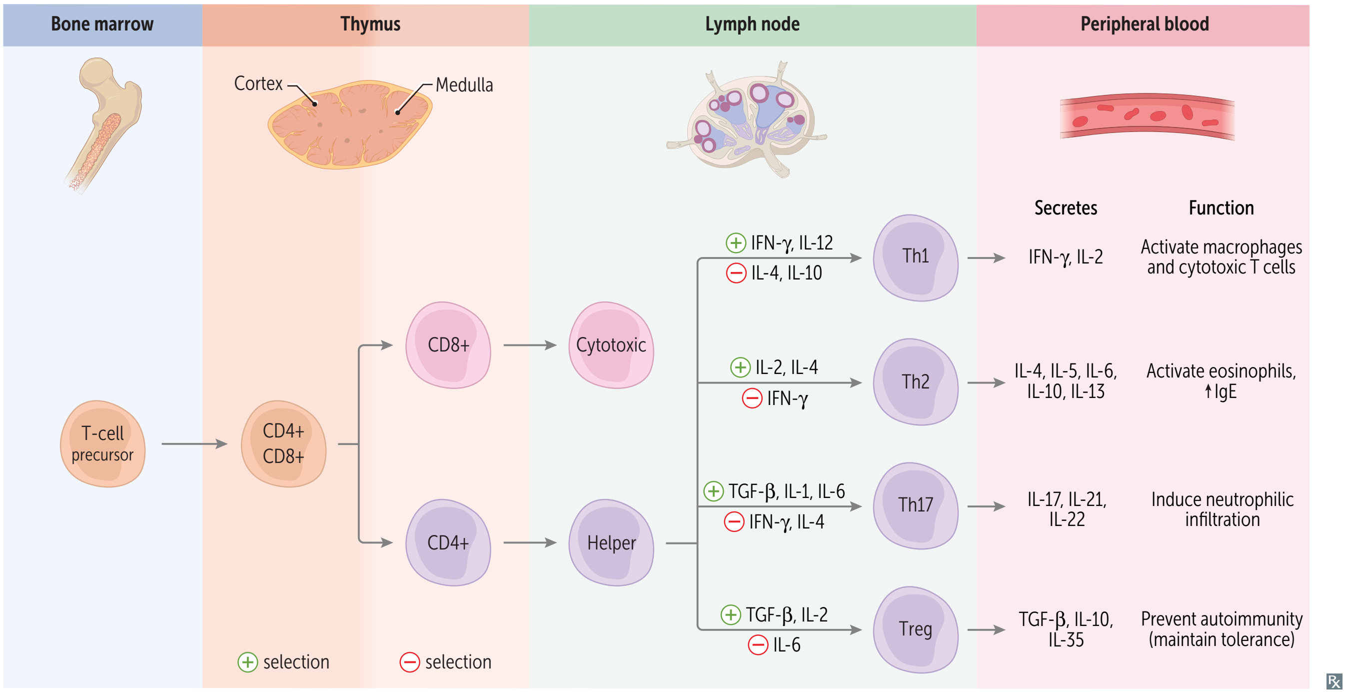
HLA SUBTYPE	DISEASE	MNEMONIC
B27	Psoriatic arthritis, Ankylosing spondylitis, IBD-associated arthritis, Reactive arthritis	PAIR
DR3	DM type 1, SLE, Graves disease, Hashimoto thyroiditis, Addison disease	DM type 1: HLA-3 and -4 (1 + 3 = 4) SL3 (SLE)
DR4	Rheumatoid arthritis, DM type 1, Addison disease	There are 4 walls in 1 "rheum" (room)

Major functions of natural killer cells	Lymphocyte member of innate immune system. Use perforin and granzymes to induce apoptosis of virally infected cells and tumor cells. Activity enhanced by IL-2, IL-12, IFN-α, and IFN-β. Produce IFN-γ → macrophage activation. Induced to kill when exposed to a nonspecific activation signal on target cell and/or to an absence of an inhibitory signal such as MHC I on target cell surface. Also kills via antibody-dependent cell-mediated cytotoxicity (CD16 binds Fc region of bound IgG, activating the NK cell).

Major functions of B and T cells

B cells	Humoral immunity. Recognize and present antigen—undergo somatic hypermutation to optimize antigen specificity. Produce antibody—differentiate into plasma cells to secrete specific immunoglobulins. Maintain immunologic memory—memory B cells persist and accelerate future response to antigen.
T cells	Cell-mediated immunity. CD4+ T cells help B cells make antibodies and produce cytokines to recruit phagocytes and activate other leukocytes. CD8+ T cells directly kill virus-infected and tumor cells via perforin and granzymes (similar to NK cells). Type IV hypersensitivity reaction. Acute and chronic cellular organ rejection.

Differentiation of T cells



Positive selection — Thymic cortex. Keeps T cells that recognize self-peptides to allow for cooperation in immune responses. Double positive thymocytes expressing TCRs that recognize self-peptide MHC complexes receive a survival signal.

Negative selection — Thymic medulla. Removes T cells that bind too strongly to self-peptides. Thymocytes expressing TCRs with high affinity for self antigens undergo apoptosis or become regulatory T cells. The autoimmune regulator (**AIR**E) protein drives negative selection, and deficiency leads to autoimmune polyendocrine syndrome (**C**hronic mucocutaneous candidiasis, **H**ypoparathyroidism, **A**drenal insufficiency, **R**ecurrent *Candida* infections). "Without **AIR**E, your body will **CHAR**".

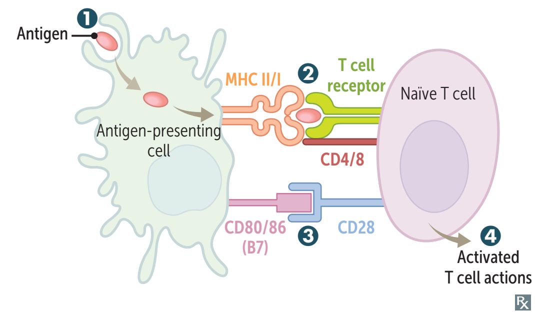
Macrophage-lymphocyte interaction — Th1 cells secrete IFN-γ, which enhances the ability of monocytes and macrophages to kill microbes they ingest. This function is also enhanced by interaction of T cell CD40L with CD40 on macrophages. Macrophages also activate lymphocytes via antigen presentation.

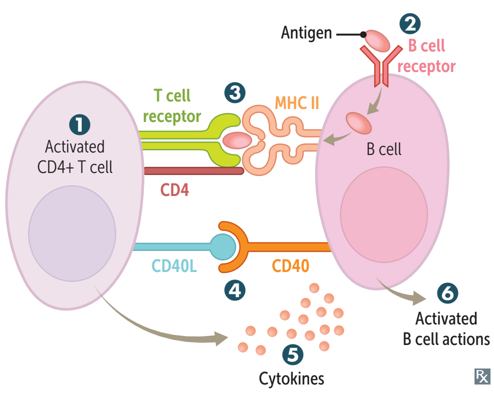
Cytotoxic T cells — Kill virus-infected, neoplastic, and donor graft cells by inducing apoptosis.
Release cytotoxic granules containing preformed proteins (eg, perforin, granzyme B).
Cytotoxic T cells have CD8, which binds to MHC I on virus-infected cells.

Regulatory T cells — Help maintain specific immune tolerance by suppressing CD4+ and CD8+ T-cell effector functions.
Identified by expression of CD3, CD4, CD25, and FOXP3.
Activated regulatory T cells (Tregs) produce anti-inflammatory cytokines (eg, IL-10, TGF-β).

IPEX (Immune dysregulation, Polyendocrinopathy, Enteropathy, X-linked) syndrome—genetic deficiency of FOXP3 → autoimmunity. Characterized by enteropathy, endocrinopathy, nail dystrophy, dermatitis, and/or other autoimmune dermatologic conditions. Associated with diabetes in male infants.

T- and B-cell activation	APCs: B cells, dendritic cells, Langerhans cells, macrophages. Two signals are required for T-cell activation, B-cell activation, and class switching.
T-cell activation	❶ APC ingests and processes antigen, then migrates to the draining lymph node. ❷ T-cell activation (signal 1): exogenous antigen is presented on MHC II and recognized by TCR on Th (CD4+) cell. Endogenous or cross-presented antigen is presented on MHC I to Tc (CD8+) cell. ❸ Proliferation and survival (signal 2): costimulatory signal via interaction of B7 protein (CD80/86) on dendritic cell and CD28 on naïve T cell. ❹ Activated Th cell produces cytokines. Tc cell able to recognize and kill virus-infected cell.
B-cell activation and class switching	❶ Th-cell activation as above. ❷ B-cell receptor–mediated endocytosis. ❸ Exogenous antigen is presented on MHC II and recognized by TCR on Th cell. ❹ CD40 receptor on B cell binds CD40 ligand (CD40L) on Th cell. ❺ Th cells secrete cytokines that determine Ig class switching of B cells. ❻ B cells are activated and produce IgM. They undergo class switching and affinity maturation.





Anergy	State during which a cell cannot become activated by exposure to its antigen. T and B cells become anergic when exposed to their antigen without costimulatory signal (signal 2). Another example of peripheral tolerance mechanism.